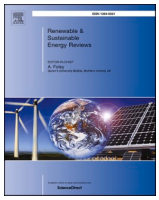
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AI-driven exploration and design of inorganic battery materials

Xuesong Wang [a], Nan He [b], Zidong Zhou [a], Jing Chen [a], Abdul Mateen [a], Yinfei Shao [a], Xiang Chen [a], Chengbin Liu [a], Altaf Mujear [a], Jianwei Zhang [a], Feng Liu [c,*], Zhihao Bao [a,**]

[a] School of Physics Science and Engineering, Tongji University, Shanghai, 200092, China
[b] School of Computer Science, Fudan University, Shanghai, 200438, China
[c] School of Computing and Information Systems, University of Melbourne, Victoria, 3010, Australia

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ABSTRACT

The increasing demand for high-performance batteries in electric vehicles, portable electronics, and renewable energy storage has spurred extensive research into novel inorganic battery materials. This review comprehensively examines the role of artificial intelligence (AI) in accelerating the discovery, design, and optimization of inorganic battery components, including cathodes, anodes, and solid-state electrolytes. We discuss various AI methodologies-such as machine learning, deep learning, transfer learning, generative models, and large language models-that integrate experimental and computational data to predict material properties and uncover structure-property relationships. Special emphasis is placed on data acquisition and standardization, model interpretability, and the challenges associated with multi-scale integration. By highlighting recent case studies and high-throughput screening strategies, the review elucidates how AI-driven approaches can overcome the limitations of traditional trial-and-error methods and high computational costs. Finally, we outline future research directions, emphasizing the potential of emerging AI techniques to further enhance material performance and accelerate the industrialization of next-generation battery technologies.

1. Introduction

In the context of global energy transition and rapid technological advancements, battery technology has become a key enabler across multiple sectors, with its importance becoming increasingly evident. As the electric vehicle market grows rapidly, portable electronic devices continue to innovate, and the demand for renewable energy storage escalates, the requirements for battery performance are becoming more stringent [1,2]. Therefore, the development of innovative battery materials and technological pathways to enhance energy density, increase power density, extend cycle life, improve safety, and reduce costs has become a critical issue that needs to be addressed [3].

Traditional battery material development processes typically rely on the "trial-and-error" method, which is inefficient and heavily dependent on the researcher's experience, resulting in underutilization of material data [4,5]. A particularly critical bottleneck in this paradigm is the prohibitive time cost of experimental validation; cycle life testing can span weeks or even months, significantly slowing iteration speed. For instance, Attia et al. demonstrated that closed-loop AI optimization could reduce the testing time for fast-charging protocols from over 500 days to just 16 days [6]. As highlighted in a recent 2025 review by Manna et al., overcoming such temporal limitations and accelerating the design-to-validation cycle has become a primary driver for integrating machine learning into materials science [7]. While computational methods, such as Density Function Theory (DFT), can provide theoretical support, their high computational cost and long run times-especially when dealing with complex battery material systems-pose significant limitations. A vast amount of data has been accumulated over the history of material development, but the data collection process lacks standardization, leading to substantial discrepancies between data obtained from different devices, environments, and researchers. Moreover, the types of experimental data vary widely, including image data, spectral data, and structural data, which complicates data integration and standardization. Data standardization is key to enabling data sharing, promoting scientific knowledge iteration, and improving data utilization efficiency.

Historically, research in materials science has been dominated by "reductionism," attempting to explain complex phenomena such as

* Corresponding author.
** Corresponding author.
E-mail addresses: fengliu.ml@gmail.com (F. Liu), zbao@tongji.edu.cn (Z. Bao).

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electronic mobility and stability with simple formulas or laws. However, materials science spans a broad range of disciplines, from microscopic electronic interactions to macroscopic electrochemical performance characterization, which often cannot be explained through a single causal relationship. Thus, blind attribution may overlook the holistic nature of complex systems, leading to misconceptions. In this context, material research should be viewed as a complex scientific system involving multi-scale, multi-level, and nonlinear interactions. Addressing high-dimensional, multi-scale data and extracting complex parameter correlations is precisely within the domain of AI algorithms based on statistics. In the field of battery material research, AI technologies are playing an increasingly critical role. Tools such as machine learning, deep learning, transfer learning, reinforcement learning, and large language models offer innovative approaches to tackling the complex challenges of battery material discovery and optimization, showcasing significant potential to revolutionize battery design and performance [8–11].

The discovery and optimization of battery materials is a highly complex and time-consuming process. Traditional methods often rely on extensive experimental trials and theoretical calculations, encountering many technological bottlenecks. However, AI technologies, through data-driven approaches, can rapidly identify potential battery materials from vast datasets, significantly improving research and development efficiency. For instance, Du et al. utilized machine learning methods to successfully screen and discover promising organic electrode molecules for lithium-ion batteries [12]. They constructed a database containing over one million organic molecules and identified 1524 potential organic electrode molecules candidates through high-throughput calculations and hierarchical clustering algorithms. Subsequently, they trained three machine learning models to predict the reduction potential of these molecules, with the support vector regression (SVR) model performing the best on the test set, yielding an R^2 value of 0.882, a mean absolute error (MAE) of 0.243, and a root mean square error (RMSE) of 0.334, indicating that the SVR model was most suitable for this task. Ultimately, the molecule Naphthalene-1,4,5,8-tetraone (NT) was selected as the positive electrode material for lithium-ion batteries, exhibiting a high reduction potential and the highest theoretical energy density, while demonstrating excellent performance in experiments, including a discharge voltage of up to 2.5 V and a remarkable cycle life of 2500 cycles at a current density of 1 Ag^{-1}. Similarly, Sheng et al. developed the BAMBOO framework, which combines a physics-inspired graph equivariant Transformer architecture, ensemble knowledge distillation, and a density alignment algorithm to achieve high-precision predictions of key properties (such as density, viscosity, and ionic conductivity) of liquid electrolytes, with an average density error of only 0.01 g/cm^3, providing an efficient molecular simulation tool for the development of liquid electrolytes for lithium-ion batteries [13].

Battery materials are typically highly complex, and their performance is influenced by multiple interacting factors. AI technologies can handle this complexity, uncovering potential relationships between material structure and performance [14]. Although traditional theoretical methods such as DFT can provide some theoretical support when dealing with complex battery materials, their high computational costs and long time requirements remain significant drawbacks. By combining AI with DFT calculations, researchers can maintain computational accuracy while greatly improving efficiency [15,16]. For example, Xu et al. have used machine learning to construct interatomic potential functions—machine learning potentials—to reveal the lithium-ion diffusion behavior in various lithium-ion conductors, including $Li_7P_3S_{11}$, $Li_{10}GeP_2S_{12}$, Li_3ErCl_6, and Li_3Yb_6 [17]. This approach effectively reduces reliance on DFT calculations, accelerating the material screening and optimization process. The application of AI in battery material research provides effective tools to address the challenges of material discovery and optimization, while also showing enormous potential to revolutionize battery design and performance. This not only holds promise for further breakthroughs in battery technology but also helps meet the growing demand for energy storage.

While battery research spans a wide spectrum of material classes, including organics and polymers, this review specifically targets inorganic battery materials. Inorganic compounds, such as layered oxides (e.g., $LiCoO_2$, $LiNixCo_{1-x}O_x$), polyanions (e.g., $LiFePO_4$), and ceramic solid-state electrolytes, differ significantly from organic counterparts in terms of stability and structural definition. While organic materials offer advantages in flexibility and molecular tunability, they currently lag in thermal stability and volumetric energy density required for high-demand applications like electric vehicles. It is crucial, however, to contextualize these two domains from a data-driven perspective. The primary distinction lies in data representation: organic materials are typically defined by discrete molecular graphs or SMILES strings, focusing on functional groups and topology; whereas inorganic materials rely on crystallographic information files (CIFs) to encode periodicity, lattice parameters, and space groups. Despite these structural differences, the underlying AI methodologies exhibit significant transferability. State-of-the-art algorithms, particularly Graph Neural Networks (GNNs) and Transformers, have proven effective in both fields. The 'message-passing' mechanisms used to learn molecular fingerprints in organics are conceptually identical to those used for inorganic crystals, provided the input embedding layers are adapted to account for periodic boundary conditions and lattice distortions. Consequently, while advancements in organic AI often inspire parallel breakthroughs in inorganic design, inorganic materials remain uniquely positioned for AI integration due to their robust data infrastructures. Standardized databases (e.g., ICSD, Materials Project) and clear structure-property relationships (e.g., crystal symmetry vs. ionic conductivity) provide a solid foundation for training high-precision models, making inorganic systems the primary frontier for demonstrating the efficacy of AI in materials science.

This review systematically examines the current landscape of AI applications in the development of inorganic battery materials, including data types and management strategies, common algorithms and their operational paradigms, representative case studies, and existing challenges. Future research directions are also explored. By analyzing the strengths and limitations of current approaches, this work aims to offer researchers a structured perspective to advance the integration of AI with inorganic materials science, thereby accelerating innovation and industrialization in battery technologies.

2. Data in inorganic battery materials

2.1. Types of data used in inorganic battery

In the field of battery research, the diversity of data types is crucial for gaining a deeper understanding of battery performance, optimizing material design, and advancing battery technology. The characteristic data of battery materials primarily include crystal structure, chemical composition, and electronic properties, all of which directly affect the overall performance of the battery.

Crystal structure is one of the key factors influencing battery performance, as it determines the pathways for ion and electron transport [18]. For instance, in lithium-ion batteries, the crystal structure of the cathode material directly affects the intercalation and de-intercalation processes of lithium ions, thereby influencing the battery's charge/discharge efficiency and cycle stability. Chemical composition is closely related to the electrochemical performance of the battery, with factors such as elemental composition and oxidation states significantly impacting the battery's voltage and capacity. Electronic properties, on the other hand, determine the material's charge transfer capability. Characteristics like electrical conductivity and band structure play a pivotal role in optimizing battery performance. By optimizing the crystal structure and chemical composition, the efficiency of ion diffusion channels can be improved, further enhancing the battery's energy density and cycle stability. Song et al. significantly enhanced the

thermal stability and electrochemical performance of the P2-Na0.67Mn0.55Ni0.30Fe0.05Cu0.10O2 cathode material under high-temperature (60 °C) and high-rate cycling conditions by optimizing the calcination process via a gradient-directed diffusion technique combined with a Cu–Fe dual-doping strategy [19]. This work demonstrates the promising potential of structural optimization and ion doping in substantially improving the performance and safety of cathode materials.

Battery performance data includes capacity, voltage, energy density, power density, and cycle stability—key metrics for evaluating battery performance. Capacity reflects the battery's energy storage capability, voltage represents the electrochemical potential difference, energy density and power density describe the battery's energy storage efficiency and charge/discharge rate capabilities, respectively, while cycle stability assesses the long-term usability of the battery. These data are essential for training machine learning models, as they help establish mapping relationships between material properties and battery performance, enabling the prediction and optimization of battery performance [20].

Experimental condition data should also be considered, which includes material synthesis parameters (such as synthesis methods, time, temperature, etc.), operating conditions (such as temperature, pressure, etc.), and electrolyte composition. These experimental conditions directly influence the material's crystal structure, chemical composition, and the final battery performance. For example, high synthesis temperatures may promote crystallinity in materials but could also lead to phase transitions or decomposition. Optimizing the electrolyte composition not only enhances the ion transport rate but also improves the stability of the battery interface, thereby significantly increasing cycle life and rate performance [21].

In conclusion, accurately acquiring and deeply analyzing these data is critical for enhancing battery performance and optimizing material design. By combining advanced data analysis methods with AI technologies, the complex interrelationships between material properties, performance, and experimental conditions can be revealed, providing strong support for the development of battery technology. Through these methods, we can effectively guide the design and optimization of battery materials, accelerating the application and innovation of battery technology across various fields.

2.2. Data collection and management

Data collection and management are crucial components in the field of materials informatics. Large-scale, high-quality materials data provide a rich and diverse resource for research, significantly advancing the continuous development of materials science. Based on the sources and characteristics of the data, materials data can be categorized into experimental data and computational data.

Experimental data are obtained through precise experimental methods to measure material properties and battery performance, and these techniques are well-established and abundant. For example, X-ray diffraction (XRD) is used to analyze the crystal structure of materials. By interpreting diffraction patterns, researchers can obtain critical information such as lattice parameters and interplanar spacing, which helps identify the type of crystal structure [15,16]. Scanning electron microscopy (SEM) and transmission electron microscopy (TEM) provide detailed microstructural characteristics of materials, including particle size, shape, distribution, and microstructure, and can also assist in analyzing the chemical composition of materials [22–24]. Electrochemical testing is a core method for evaluating battery performance. By measuring charge and discharge curves, key performance indicators such as capacity, charge/discharge efficiency, and cycle life can be precisely obtained. These techniques complement each other, providing comprehensive information to support material research and optimization. However, accurate and reproducible experimental data collection still faces many challenges, including fluctuations in experimental conditions, variations in sample preparation, and inconsistencies in equipment and personnel operations, all of which may affect data accuracy and reproducibility [25,26].

In materials research, computational data are primarily generated through first-principles calculations, molecular dynamics simulations, and Monte Carlo methods [27]. First-principles calculations, based on quantum mechanics, predict the electronic structure of materials at the electronic level, such as electronic density of states, band structure, and thermodynamic properties like formation energy, enthalpy change, and entropy change [28]. Molecular dynamics simulations focus on studying the dynamic behavior of materials at the atomic scale [29]. By simulating atomic motion trajectories under specific time and temperature conditions, these simulations reveal atomic diffusion mechanisms, phase transitions, and mechanical properties. Monte Carlo methods are primarily used to address random problems in complex systems, such as simulating defect distribution and analyzing phase transition dynamics [30]. Computational data provide valuable support for microscopic information that is difficult to obtain experimentally and can be used for performance prediction and experimental design, thus improving experimental efficiency. However, computational data also face limitations, such as the accuracy of model assumptions and parameter selections, as well as the high computational cost. Furthermore, despite the abundance of structural data in databases like ICSD, high-quality labeled experimental data for specific properties (e.g., ionic conductivity, cycle life) remains significantly sparse for inorganic systems compared to the vast chemical space. This data sparsity necessitates the adoption of data-efficient acquisition strategies, such as Active Learning, to iteratively guide experiments toward the most informative candidates.

Materials databases integrate large amounts of experimental and computational data, providing convenient data querying and analysis platforms for researchers [31–36]. In Table 1, we have summarized the similarities and differences among the current major materials

Table 1
Common material databases, including data types, data sources, and data scale.

Database	Main Content	Data Source	Data Size
Materials Project [31]	Crystal structure, thermodynamics, electronic structure, elasticity, etc.	High-throughput calculations, some experimental data	Over 800,000
ICSD (Inorganic Crystal Structure Database) [32]	Inorganic crystal structure data, covering various elements and compounds	Experimental data (X-ray diffraction, electron diffraction, etc.)	Over 210,000
CSD (Cambridge Structural Database)	Crystal structure data of organic and metal-organic compounds	Experimental data (X-ray diffraction)	Over 1.25milion
AFLOW [33]	Structure, energy, thermodynamics, electronic and elastic properties of inorganic compounds	High-throughput calculations (DFT), some experimental validation data	Over 700milion
Springer Materials	Physical, chemical, and thermodynamic data of materials (metals, alloys, ceramics, etc.)	Springer publications, Landolt-Börnstein, literature, experimental data (primarily)	Over 290,000
Open Quantum Materials Database (OQMD) [34,35]	Structure, formation enthalpy, electronic structure, etc. for inorganic materials	High-throughput calculations (DFT)	Over 850,000
Atomly [36]	Physical, chemical, and electrochemical properties of materials, supporting machine learning predictions	Integrated data from open databases, experimental data, high-throughput calculations	Over 760,000

databases. Common materials databases, such as the Inorganic Crystal Structure Database (ICSD), Materials Project, and AFLOW, contain rich information about material structures, properties, and synthesis methods, offering significant support for material discovery, design, and optimization. For example, researchers can use the ICSD database to search for specific materials' crystal structures, obtaining detailed information such as atomic arrangement and lattice parameters, which can help infer physical and chemical properties like crystal symmetry, density, and hardness. The Materials Project database provides a wealth of computational data, including electronic structure and thermodynamic properties, helping researchers predict and analyze material performance. Additionally, materials databases play an important role in data mining and machine learning. By deeply analyzing the vast amounts of data in these databases, researchers can uncover complex relationships between material properties and structure, providing crucial insights for the development of new materials. However, materials databases still face challenges related to data accuracy, completeness, and timely updates. For example, experimental data may contain errors due to variations in experimental conditions and operations, and data from first-principles calculations may have systematic deviations due to limitations in calculation precision and methods. Furthermore, as materials science advances rapidly, new materials are constantly emerging, but some data on novel materials may not be promptly included, which could impact the comprehensiveness and reliability of research.

Therefore, continuously improving and updating materials databases is a key task in the development of materials informatics. To ensure data accuracy, effective data review mechanisms need to be established. Moreover, strengthening data collection, improving the completeness of databases, and staying current with the latest developments in materials science are essential to ensure timely updates. These measures will not only enhance the quality and reliability of the databases but also provide researchers with powerful tools, thus driving the rapid development of materials science.

3. AI algorithms and their work paradigms in the battery field

AI was formally born at the Dartmouth Conference in 1956. Initiated by John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon, this conference gathered scholars to discuss the simulation of human intelligence by machines, officially coining the term "Artificial Intelligence." In the nearly 70 years since, numerous classic algorithms have emerged, as shown in Fig. 1 [37]. Its development can be conceptually divided into three stages: (1) Symbolic AI (Knowledge--Driven): Originating in the 1950s, Alan Turing proposed theoretical frameworks for testing machine intelligence, while John McCarthy invented LISP in 1958. This paradigm established rule-based AI research through symbolic manipulation for knowledge representation and logical reasoning. (2) Deep Learning (Data-Driven): Emerging in the early 21st century with enhanced computing power and big data, breakthroughs like Geoffrey Hinton's deep neural networks (2006) and Yann LeCun's 1990s Convolutional Neural Networks (CNN) for image recognition revolutionized perceptual intelligence. Models such as LSTM automated feature extraction from massive datasets, advancing data-driven cognition. (3) Third-Generation AI: Recent efforts integrate knowledge-driven and data-driven approaches, combining the interpretability and logic of symbolic systems with deep learning's feature extraction. By embedding prior knowledge, these hybrid models improve generalization capabilities, aiming to build more reliable, adaptive, and explainable AI systems [38]. In the materials field, due to data scarcity and the need for model interpretability, there is a general preference for algorithms with clear structures that can easily integrate materials science prior knowledge with data. However, with the development of materials databases and the advancement of AI technologies, increasingly complex models have emerged. Traditional machine learning, deep learning, reinforcement learning, and transfer learning are among the most commonly used AI algorithms in the battery field.

3.1. Working paradigms of machine learning/deep learning

Machine learning is currently the most prominent branch of AI, with deep learning being an important subfield of machine learning. Fig. 2 illustrates a comprehensive workflow that encapsulates the key stages involved in employing machine learning techniques for the discovery and design of battery materials. Although traditional machine learning and deep learning differ slightly in their workflows, they can generally be divided into the following key steps, each involving clear tasks and methods to ensure the performance and practical utility of the model.

3.1.1. Data preparation

In data-driven materials science research, the primary task is to define the problem context, objectives, and evaluation metrics. Data sources encompass experimental measurements (e.g., battery material testing), computational simulations (density functional theory (DFT), molecular dynamics), specialized databases (Materials Project, Open Quantum Materials Database (OQMD)), and published literature. Data quality must rigorously mitigate noise and annotation errors, while domain-specific prior knowledge—particularly from physics and chemistry—enhances data utility. For instance, predictive models for battery materials often require manually engineered features, such as crystallographic symmetry (influencing structural stability), bond length distributions (linked to mechanical strength), and electronegativity (governing charge transfer)—key physicochemical descriptors critical for performance prediction. Preprocessing involves handling missing values and applying normalization/standardization to ensure feature comparability. Datasets are typically partitioned into training, validation, and test sets at an 8:1:1 ratio, with data augmentation techniques (e.g., noise injection or interpolation) employed when necessary. This systematic framework integrates data-driven pattern recognition with domain expertise to achieve robust and interpretable predictive capabilities, aligning with the demands of advanced materials discovery and optimization.

3.1.2. Model selection

Model selection needs to be considered comprehensively, taking into account the nature of the problem, the characteristics of the data, and practical needs. The first step is to clarify the nature of the task according to the problem's objective. For instance, classification tasks are suitable for material category recognition or state prediction, regression tasks are used for predicting continuous values such as battery capacity or lifetime, generative tasks can generate new material structures or optimize formulas, and optimization tasks are used for experimental design or path optimization. In addition to considering the task, data characteristics must also be analyzed. The first step is to assess the data scale. Small sample or low-dimensional data is suitable for traditional machine learning algorithms, such as support vector machines or random forests, while large-scale or high-dimensional data is more suited to deep learning. Secondly, data type matters—structured data, such as battery performance indicators, is ideal for traditional machine learning, while unstructured data, such as micrographs or crystal structure data, is better suited to deep learning models (e.g., CNN, GNN). If the data contains time-series characteristics, such as charge-discharge curves, recurrent neural networks (RNN) or Transformers are more appropriate.

3.1.3. Model training and optimization

The loss function is used to quantify the discrepancy between the model's predictions and the true values, serving as the core driving force for parameter optimization. In machine learning and deep learning, the loss function guides the parameter updates by quantifying prediction errors, significantly affecting model performance. Common loss functions include mean square error (MSE), MAE, Cross-Entropy Loss, and

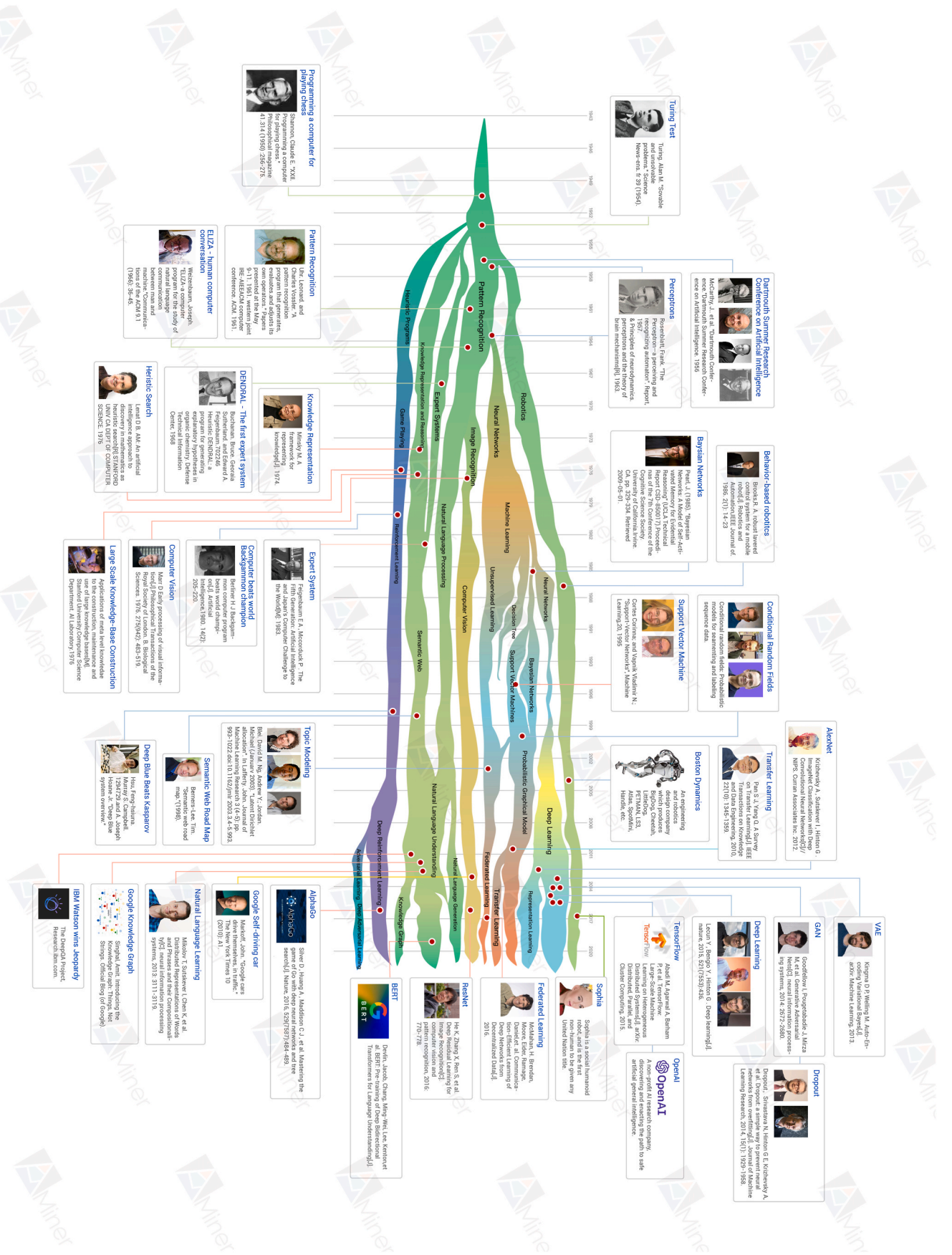


Fig. 1. The History of AI. Some key scholars, papers, events, technologies, fields, time points, applications, and the relationships among them. Reproduced with permission [37]. Copyright 2008, Association for Computing Machinery (ACM).

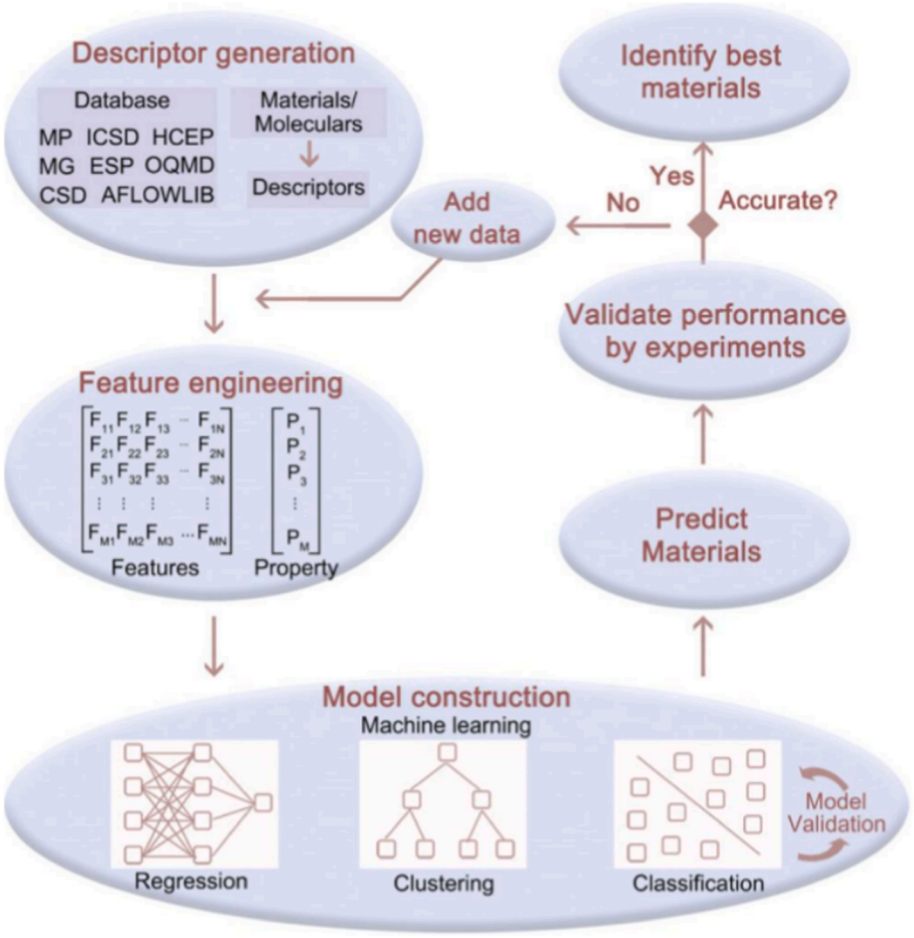


Fig. 2. An essential process for discovering and designing battery materials using machine learning methods [10]. Copyright 2022, Advanced Materials.

0–1 Loss. Among these, Cross-Entropy Loss is widely used in classification tasks due to its sensitivity to the differences in probability distributions. Its mathematical expression is given by:

$$L_{CE} = -\frac{1}{n}\sum_{i=1}^{n}\sum_{c=1}^{C} y_{i,c}\, log\left(\hat{y}_{i,c}\right) \tag{1}$$

where C is the number of classes, $y_{i,c}$ represents the true label (0 or 1) for sample i in class C and $\hat{y}_{i,c}$ is the predicted probability. Research has shown that the design of customized loss functions can significantly enhance model performance. For example, Yang et al. proposed an uncertainty-aware loss function for crystal structure relaxation prediction, which effectively combines atomic distance loss with displacement loss to capture uncertainties in both data and models, thus improving prediction robustness [39].

The optimization algorithm is the essential tool for parameter adjustment during training. Stochastic Gradient Descent (SGD), as a fundamental algorithm, updates parameters by randomly selecting small batches of data to compute the gradients, providing high computational efficiency, although its convergence can be susceptible to gradient fluctuations. To enhance convergence stability, adaptive learning rate optimizers, such as Adam, are widely adopted. Adam combines the advantages of AdaGrad and RMSProp by dynamically adjusting the parameter update step based on the first moment (momentum term) and the second moment (adaptive learning rate) of the gradients, accelerating convergence while mitigating oscillations [40]. Additionally, Momentum methods, which incorporate a weighted average of past gradient directions, smoothen the parameter update trajectory and are well-suited for non-stationary or high-curvature loss surfaces [41].

Regularization techniques are critical strategies for preventing model overfitting. Overfitting manifests when a model performs excellently on training data but shows poor generalization on test data. Traditional regularization methods, such as L_2 regularization, add a penalty term based on the squared parameters to the loss function, constraining parameter magnitudes and preventing excessive model complexity [42]. Stochastic regularization techniques, such as Dropout, randomly omit neurons during training, forcing the network to learn redundant representations, thereby enhancing generalization [43]. Batch Normalization (BN) further alleviates internal covariate shift by standardizing the input distribution of each layer, while also implicitly acting as a regularizer.

Learning rate scheduling and parameter initialization also influence training efficiency. Learning rate decay (e.g., cosine annealing) gradually reduces the step size to balance convergence speed and accuracy, while warmup strategies slowly increase the learning rate during the initial phase to avoid unstable gradient directions [44]. Parameter initialization methods, such as Xavier and He initialization, adjust the initial weight distribution to ensure consistent variance of signals during both forward and backward propagation, thereby accelerating convergence [45].

In summary, effective model training requires a comprehensive approach, integrating loss function design, optimization algorithm selection, regularization strategies, and learning rate scheduling, alongside domain-specific prior knowledge (such as Yang et al.'s uncertainty-aware loss function), to achieve optimal performance. Future research could further explore the synergy between automated hyperparameter optimization techniques (such as Bayesian optimization) and model

ensembles (e.g., Stacking) for complex tasks.

3.1.4. Model evaluation and interpretation

Model performance is assessed using metrics such as R^2, MSE, or accuracy. The coefficient of determination R^2 measures how well the model fits the data, with values ranging from 0 to 1, where values closer to 1 indicate better explanatory power of the model. MSE directly reflects the average error between the predicted and true values; smaller values indicate higher prediction accuracy. Accuracy, on the other hand, is used for classification tasks and measures the proportion of correctly classified samples.

To interpret the important features of the model, methods like SHAP values (Shapley Additive exPlanations), or LIME (Local Interpretable Model-agnostic Explanations) can be employed. SHAP values, based on cooperative game theory, assign an importance value to each feature and help explain the relationship between model predictions and features. LIME approximates the model locally to explain its predictions. It allows interpretation of individual predictions without depending on the model's structure, helping researchers understand the decision-making process of the model and identify the key features that significantly influence the model's predictions.

3.2. Common AI algorithms used in the battery field

This review does not aim to provide an exhaustive survey of all artificial intelligence algorithms; rather, it focuses on those that are widely applied or demonstrate significant potential within the domain of battery research. Based on their operational paradigms, computational complexity, and data processing capabilities, these algorithms can be categorized into five distinct classes: (1) Traditional Machine Learning, (2) Neural Networks and Graph Learning, (3) Generative Models, (4) Transfer Learning, and (5) Large Language Models. In this context, "Traditional Machine Learning" refers to classical algorithms established prior to the widespread adoption of deep learning, such as regression models, decision trees, and support vector machines (SVM). Grounded in statistical learning theory, these methods are primarily characterized by their reliance on manually engineered features to represent material properties. They feature simple, transparent model structures that offer high interpretability and are well-suited for small-scale datasets. However, traditional machine learning lacks automatic feature extraction capabilities and struggles to directly process unstructured data. The subsequent categories represent an evolution toward deep representation learning, autonomous structure generation, and knowledge integration, enabling the handling of more complex, high-dimensional data modalities.

3.2.1. Traditional machine learning algorithms

Traditional machine learning algorithms are a class of tools based on mathematical statistical models, widely used in classification, regression, and clustering tasks [46,47]. They can be categorized into supervised learning and unsupervised learning. The main characteristics of these algorithms include their relatively low data requirements, a strong focus on data feature extraction and selection, high computational efficiency, and good interpretability. Common traditional machine learning algorithms include linear regression, logistic regression, support vector machines (SVM), decision trees, random forests, k-nearest neighbors (KNN), and clustering algorithms (such as K-means). Although traditional machine learning algorithms show significant advantages in the aforementioned tasks, they have limited capability in processing high-dimensional data, rely heavily on manual feature engineering, and are difficult to apply to unstructured data, such as images and complex molecular structures. In contrast, deep learning can handle more complex data types through automatic feature extraction, but its computational cost is high and its interpretability is poor. In practical applications, traditional machine learning algorithms remain an efficient and reliable tool in battery material research, especially in scenarios with limited data and the need to quickly construct models.

3.2.2. Neural networks and graph

Neural network algorithms are computational models that simulate the structure and function of the human nervous system and are widely applied in materials [48]. Common neural network algorithms include feedforward neural networks (FNN), convolutional neural networks (CNN), and long short-term memory networks (LSTM). In battery material research, the graph is a particularly suitable method for describing and analyzing complex material structures. In materials science, the crystal structure of materials can be naturally represented as a graph. Fig. 3 illustrates the utilizing of different graph structures to represent materials, where nodes represent atoms and edges represent chemical bonds. As an extension of traditional neural networks, graph neural networks (GNNs) are designed to handle graph-structured data. These graph-based models are particularly well-suited for inorganic materials as they can naturally capture the periodicity of crystal lattices and encode topological invariance. The Crystal Graph Convolutional Neural Networks (CGCNNs) represent crystal structures as graphs and extract features through convolution operations. CGCNNs can directly learn material properties from crystal structure data without requiring manually constructed features. This approach offers several distinct advantages: First, it can efficiently predict material properties, such as the Gibbs free energy and electrochemical potential of zinc-ion battery materials [49]. Second, CGCNN can provide chemical insights for material design by extracting features from the local chemical environment. For example, by analyzing the energy contributions of different atomic positions in a perovskite structure, CGCNN can reveal more stable structural combinations. To address diverse challenges in graph-structured data, researchers have developed multiple GNN variants, such as Atomistic Line Graph Neural Network (ALIGNN) [50], Orbital graph convolutional neural network (OGCNN) [51], Graph Variational Auto Encoder (GraphVAE) [52].

3.2.3. Generative Models

Generative Adversarial Networks (GANs): Generating material structures with specific properties. The fundamental idea of GANs is the adversarial game between the generator and the discriminator. The generator attempts to produce samples that resemble the true data distribution, while the discriminator strives to distinguish between the generated data and the real data. During the training process, the generator and discriminator continually optimize, making the generated data more and more realistic. Nouira et al. introduced CrystalGAN, a generative adversarial network (GAN) designed to generate complex crystal structures [54]. As shown in Fig. 4, the architecture of CrystalGAN can be divided into two stages: the first stage performs cross-domain pseudo-binary sample generation, while the second stage conducts ternary structure synthesis under geometric constraints. Through phased adversarial training and the deep integration of geometric constraints, CrystalGAN successfully achieves high-complexity generation of crystal structures from binary to ternary systems. It was used to discover ternary stable compounds for hydrogen storage materials. In battery materials research, GANs generate high-energy-density, long-cycle-life structures by tuning generator noise and optimizing training, enabling exploration of diverse structural spaces and novel approaches for high-performance material discovery.

Variational Autoencoders (VAE): Learning latent distributions of materials for generating new materials. VAE consists of an encoder and a decoder, where the encoder maps input data to the latent space and learns the latent distribution of the data, while the decoder generates reconstructed data based on the representation in the latent space. The first term in the loss function measures the difference between the reconstructed data and the original data, while the second term constrains the latent distribution to be similar to a prior distribution. Through training the VAE, latent feature representations of material data can be learned, which can then be used to generate new materials

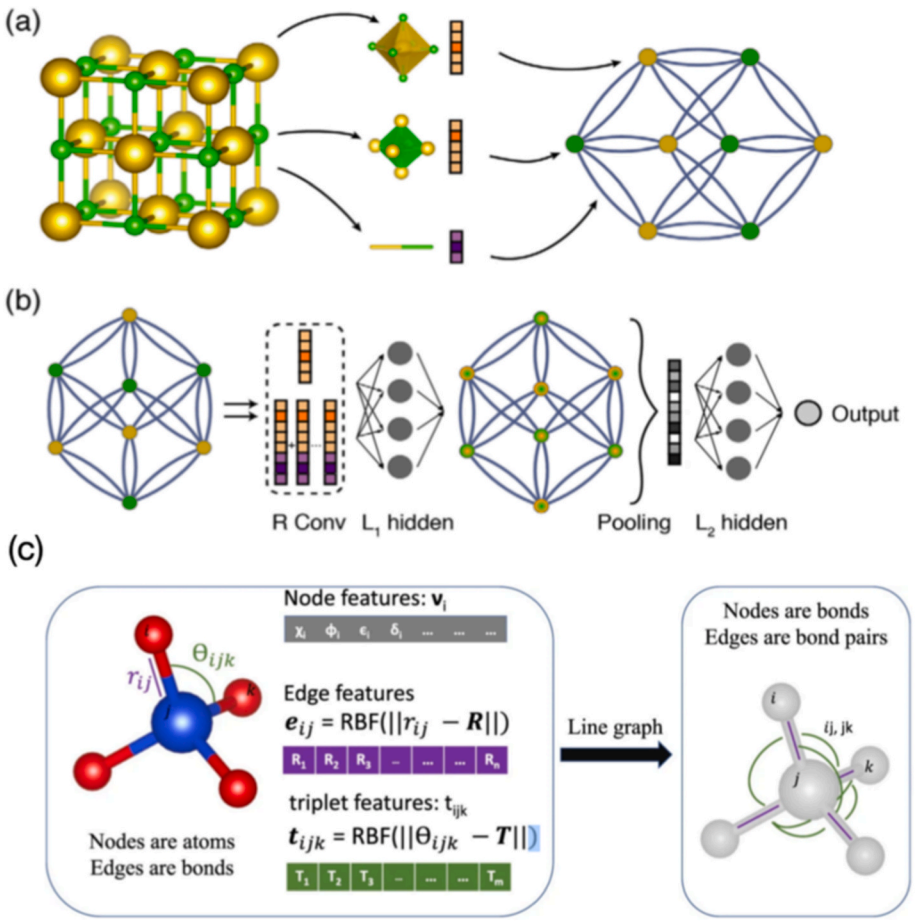


Fig. 3. The utilizing of diverse graph structures to represent materials. (a) Construction of the crystal graph. The crystal is transformed into a graph, where the nodes represent atoms in the unit cell, and the edges represent atomic connections. The nodes and edges are represented by corresponding vectors of atoms and bonds in the crystal [53]. Copyright 2018, Physical Review Letters (b) Convolutional neural network structure at the top of the crystal graph. R convolutional layers and L_1 hidden layers are constructed at each node to form a new graph, where each node represents the local environment of each atom. After pooling, a vector representing the entire crystal is connected to L_2 hidden layers, followed by an output layer to provide the prediction [53]. Copyright 2018, Physical Review Letters (c) Undirected crystal graph representation and corresponding line graph construction for a SiO_4 polyhedron [50]. Copyright 2021, npj Computational Materials.

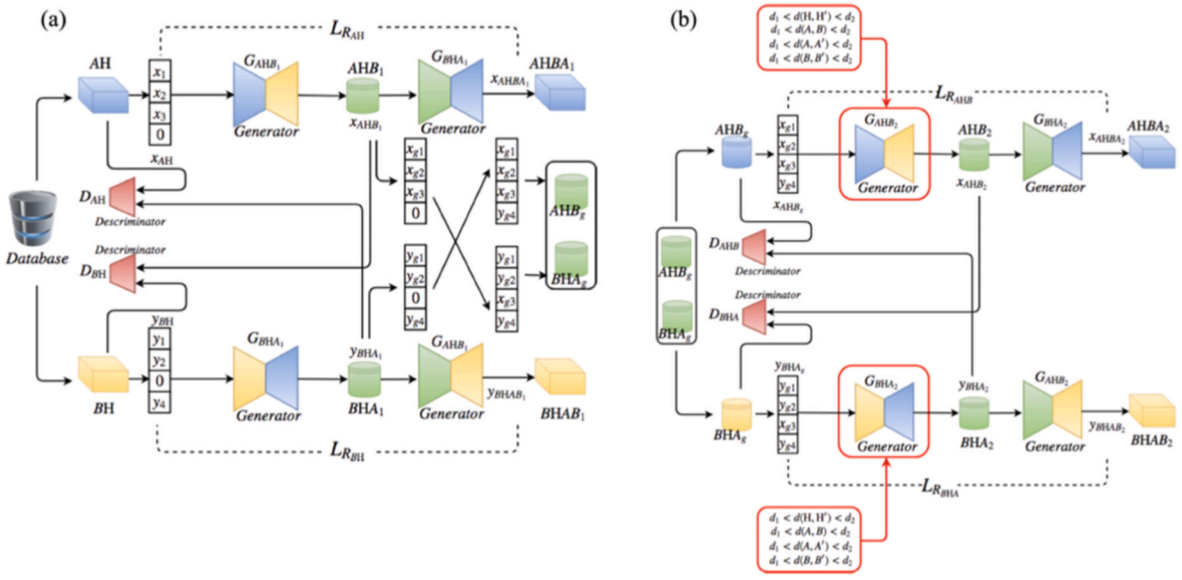


Fig. 4. The CrystalGAN architecture. (a) First step of CrystalGAN; (b) Second step of CrystalGAN [54]. Copyright 2019, Association for the Advancement of Artificial Intelligence (AAAI).

with similar characteristics. In battery material research, VAE can help researchers discover new material structures, explore the potential relationships between material performance and structure, and provide support for innovative design in battery materials.

Diffusion Models: A generative approach utilizing probabilistic diffusion processes, comprising a forward phase (data perturbation into noise) and a reverse phase (data reconstruction). These models learn data distribution gradients to guide generated samples toward high-density regions. In materials science, they enable the design of complex structures by incorporating physical constraints (e.g., periodicity, stability, bonding). For example, Xie et al. developed a generative framework, Crystal Diffusion Variational Autoencoder (CDVAE), based on diffusion models and VAEs, specifically designed for generating periodic stable material structures. CDVAE integrates a Noise-Conditional Score Network (NCSN) into the VAE decoder to optimize material structures [55]. Fig. 5 compares the algorithmic processes of CDVAE and Con-CDVAE [56]. Con-CDVAE is an extended version of CDVAE. By means of multi-attribute handling, a dual diffusion model architecture, two-step training, and generation strategy optimization, it addresses the limitations of CDVAE in terms of attribute diversity and conditional generation flexibility, making it suitable for more complex crystal design tasks (such as the generation of materials with specified band gaps and formation energies). Its core contribution lies in expanding conditional generation from the scenario of single continuous attributes to that of multiple attributes, and verifying the cross-conditional and cross-dataset generation capabilities through experiments.

3.2.4. Transfer Learning

The fundamental concept of transfer learning is to utilize a pre-trained model from one task and transfer its knowledge to another related task. In the field of batteries, acquiring large-scale labeled datasets is often costly, making transfer learning an effective approach. This is particularly relevant for inorganic materials, where experimental datasets are often too sparse to train deep models from scratch. Transfer learning overcomes this by pre-training on large-scale computational databases (e.g., Materials Project) and fine-tuning on limited experimental data, effectively bridging the gap between abundant theoretical data and scarce experimental ground truth. It leverages pre-trained models from other domains, such as image processing and natural language processing, and fine-tunes the model parameters to adapt them to battery material research tasks. For instance, a convolutional neural network (CNN) pre-trained on ImageNet has learned rich image feature representations. By transferring this knowledge to tasks such as analyzing microscopy images of battery materials, it can reduce training time and data requirements, enabling the rapid development of effective models. By constraining the fine-tuned parameters to remain close to those of the pre-trained model, overfitting to new data during fine-tuning can be avoided, while maintaining the generalization capabilities of the pre-trained model, thus enhancing performance in battery-related tasks.

3.2.5. Large language Model (LLM)

In recent years, large models in the field of AI have been developing rapidly, giving rise to numerous remarkable achievements [57–60]. In inorganic materials research, LLMs have demonstrated significant potential in aspects such as information processing, experimental design, and system integration. Firstly, LLMs accelerate literature mining by extracting structured data from a vast amount of literature. For example, the ChatGPT chemistry assistant reduces the literature processing time by 67 % and achieves an accuracy rate of 87 % in predicting the crystallinity of metal-organic frameworks (MOFs) [61,62]. Fig. 6 illustrates the detailed workflow of the ChatGPT Chemistry Assistant. The assistant extracts synthesis conditions from MOF research papers through three processes—manual preselection of paragraphs for extraction, automatic

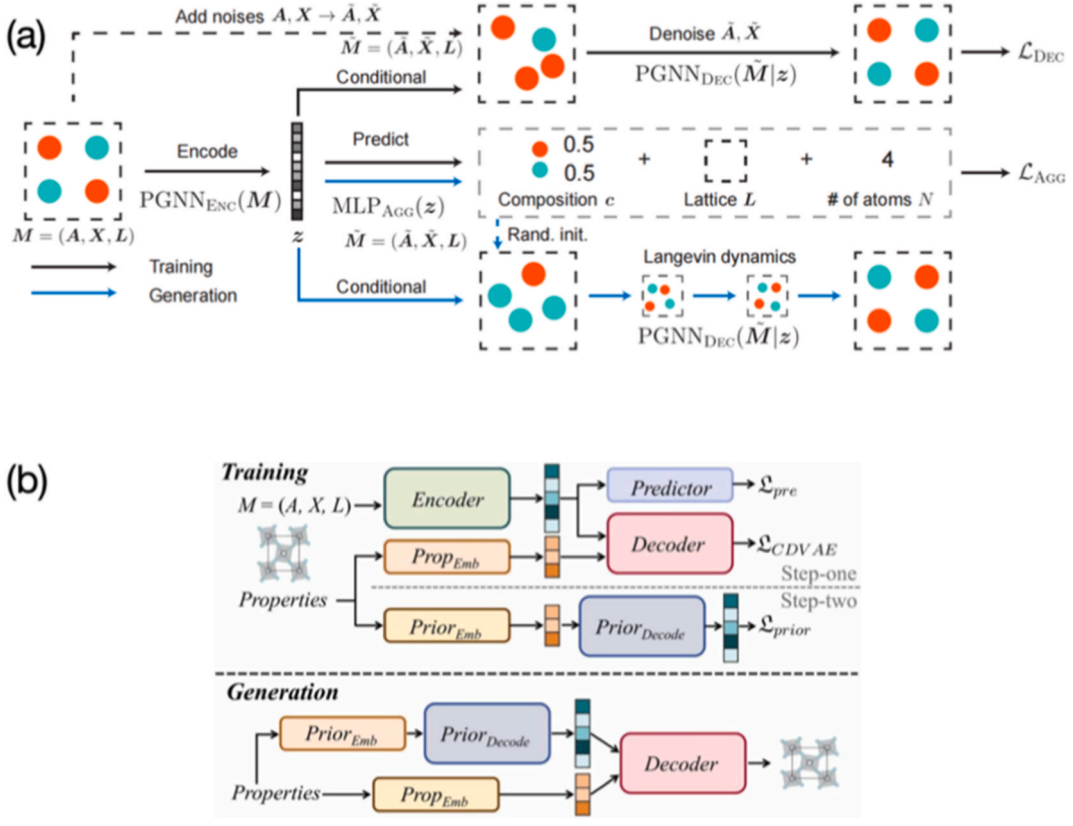


Fig. 5. (a) Overview of the proposed CDVAE approach [55]. Copyright 2022, International Conference on Learning Representations (ICLR) (b) Training and generation flow chart of Con-CDVAE [56]. Copyright 2024, Computational Materials Today.

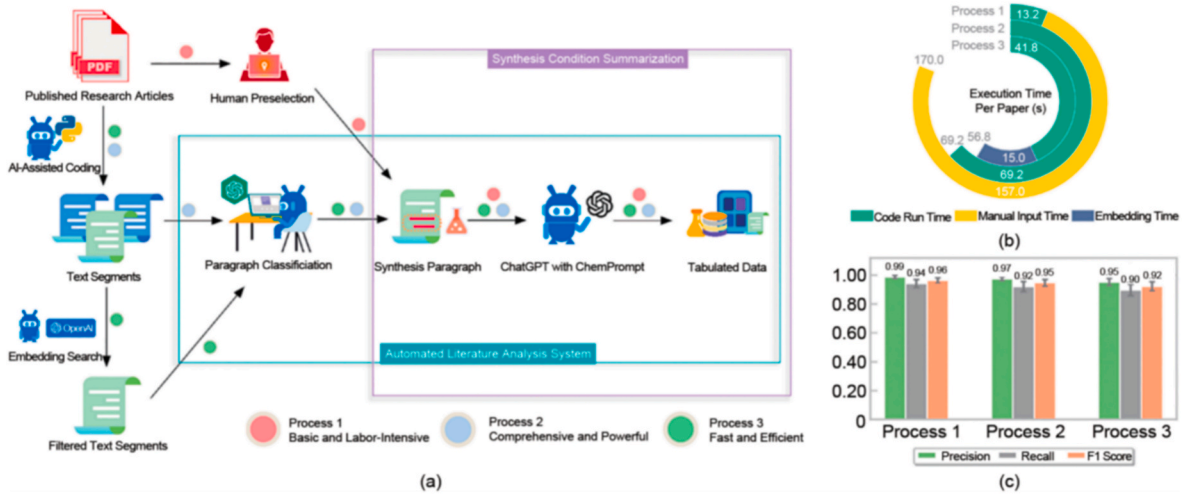


Fig. 6. Schematics of the ChatGPT Chemistry Assistant. (a) ChatGPT Chemistry Assistant workflow, each process is distinctively labeled with red, blue, and green dots; (b) Comparison of the average execution time required by each process to read and process a single paper; (c) Aggregate average precision, recall, and F1scores for each process. Standard deviations are represented by grey error bars in the chart [61]. Copyright 2023, Journal of the American Chemical Society.

classification of experimental sections in full texts, and rapid filtration of irrelevant content—incorporating three prompt engineering principles (minimizing hallucination, detailed instructions, and structured output) to ensure accurate data structuring and provide support for subsequent analysis and model training. Similarly, frameworks like MoIReGPT have achieved the mutual conversion between molecular descriptions and structural characterizations, simplifying the data collation process [63]. LLMs also drive hypothesis generation and autonomous experiments. The "machine chemist" system developed by the University of Science and Technology of China screened 4865 molecules from 15,979 papers and optimized the performance of luminescent materials [64]. The A-Lab platform at the University of California, Berkeley combined LLMs with a robotic system and synthesized 41 novel inorganic compounds within 17 days with a success rate of 71 % [65]. The reinforcement of LLMs can also synergize with experimental workflows. For example, the Coscientist platform at Carnegie Mellon University integrated LLMs with

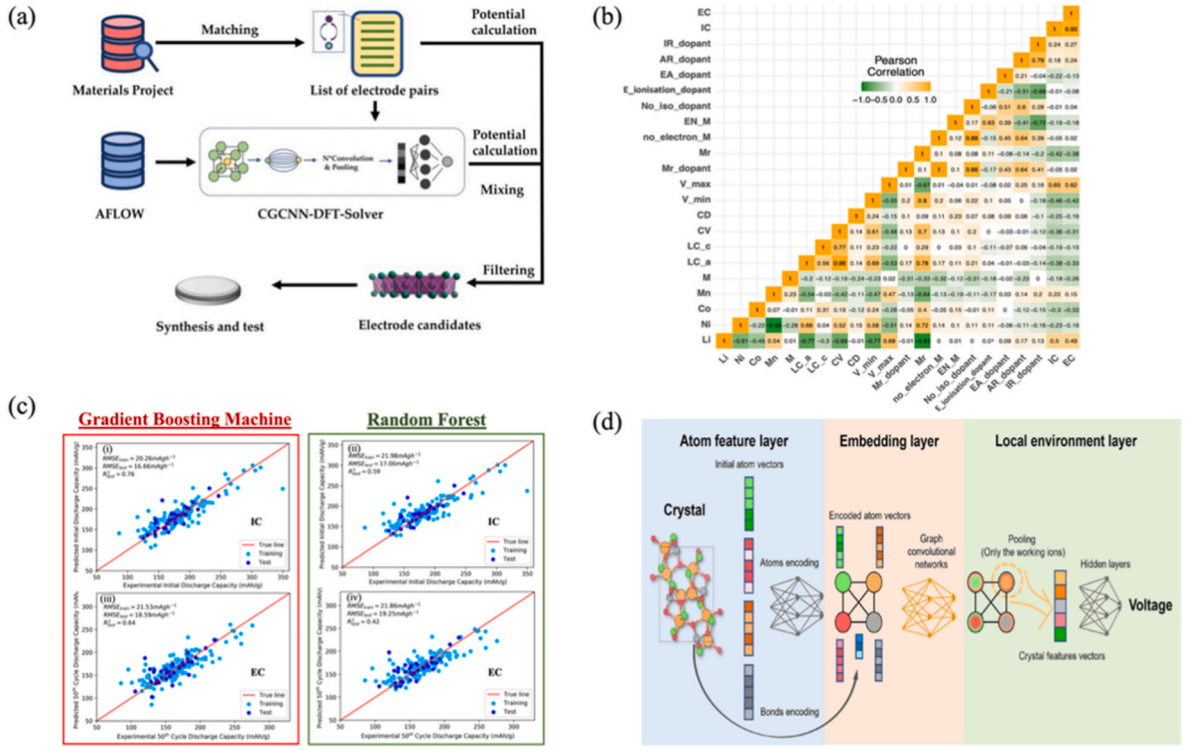


Fig. 7. (a) Schematics of the ML-assisted battery electrode material screening protocol [49]. Copyright 2021, npj Computational Materials (b) Results matrix of Pearson coefficient correlations for every pair of variables in the data set, including covariate variables, Li, Ni, Co, Mn, M, LC_a, LC_c, CV, V_min, V_max, CD, Mr, Mr_dopant, No_electron_M, EA_dopant, No_iso_dopant, AR_dopant, IR_dopant, E_ionisation_dopant, and EN_M, and two response variables, IC and EC. The estimated correlation values are distributed within the range of −1 to +1, with a number reaching either end value implying a more perfect negative correlation and positive correlation, respectively [71]. Copyright 2021, American Chemical Society (c) Scatter plots comparing experimental IC and FC to predicted IC and FC computed by GB and RF models for doped NCM cathode systems [71]. Copyright 2021, American Chemical Society (d) Illustration of the modified CGCNN model for voltage prediction [72]. Copyright 2025, Elsevier B.V.

an automated laboratory to design and execute complex chemical reactions, demonstrating chemical reasoning capabilities [66]. Despite progress, challenges remain in developing high-quality domain-specific datasets, balancing model generalization and accuracy, and meeting computational demands. Nevertheless, integrating LLMs with inorganic materials science holds promise for reshaping discovery paradigms and enhancing efficiency in tackling global energy and sustainability challenges.

4. Application of AI in inorganic battery materials

4.1. Cathode materials

Over the past three decades, with advancements in technology, the energy density of rechargeable batteries has increased from 150 Wh/kg in the 1990s to over 300 Wh/kg in the 2020s, leading to their widespread application [67]. However, there remains a need to develop new electrode materials that are either lower in cost or at least comparable in performance to current commercialized electrode materials. The development and application of these materials have seen exponential growth in both experimental and computational domains [68]. Such a vast demand cannot be fully supported by existing experimental methods and first-principles calculations. Initial efforts focused on high-throughput screening to identify candidates within vast databases. For instance, Zhou et al. developed a machine learning tool based on Crystal Graph Convolutional Neural Networks (CGCNN) [49]. Fig. 7 (a) shows the Schematics of the ML-assisted battery electrode material screening protocol. By screening approximately 130,000 inorganic materials from the Materials Project (MP) and AFLOW databases, combining database calculations and machine learning-predicted voltage data, they identified around 80 potential cathode materials for zinc-ion batteries with high voltage (>0.5 V vs Zn/Zn^{2+}) and high capacity (>100 mAh g^{-1}), of which approximately 10 have been experimentally discovered and about 70 are new potential candidates. The predicted voltages showed good agreement with experimental data, and the CGCNN revealed the deep impact of local chemical environments on battery performance, providing chemical insights for optimizing cathode materials.

Similar high-throughput screening studies can be applied to the development of cathode materials for different types of metal-ion batteries [69]. These studies can offer valuable strategies for the development of novel batteries. For example, Moses et al. proposed a deep neural network (DNN) model trained on the Materials Project database to predict the average voltage and volume changes during the charge-discharge process of metal-ion battery electrode materials [70]. Despite the abundance of sodium (Na) and potassium (K) in the Earth's crust, they are rarely used as commercial electrode materials, as they typically do not form stable or feasible compounds with common lithium-ion battery electrode materials. This study explored the potential of replacing Li-ion with Na-ion in novel electrode materials. They extracted 306 unique electrode structures and chemical composition features from three self-constructed datasets. Principal component analysis (PCA) was then used to identify important features, leading to the final selection of 106 features. Two separate DNNs were trained to predict the average voltage (Vav) and volume change percentage (% vol. change). All models consisted of three hidden layers with 360, 90, and 30 nodes, respectively, using MSE as the loss function and evaluated by 10-fold cross-validation. For the Vav model, the final MAE on the test set was 0.35 V (for the entire ion dataset), 0.45 V (for the alkali metal dataset), and 0.43 V (for the lithium-ion dataset). For the volume change percentage model, the final MAE on the test set was 10.6 % (for the entire ion dataset), 7.8 % (for the alkali metal dataset), and 3.8 % (for the lithium-ion dataset). However, the MAE values and their standard deviations were high, which affected the prediction of volume changes for novel electrode materials. This phenomenon was attributed to the severe imbalance in the dataset, where approximately 80 % of the volume change percentage values were below 10 %, while the data range extended from 0 % to 26,000 %. By removing data instances with volume changes greater than 30 %, significant improvement was observed, with the MAE reduced to 2.7 % (for the entire ion dataset), 2.4 % (for the alkali metal dataset), and 2.1 % (for the lithium-ion dataset). This study demonstrates that appropriate data processing techniques can significantly enhance the performance of machine learning models. Although the models are not yet perfect, they provide an efficient method for the preliminary high-throughput screening of novel metal-ion electrode materials.

Building on screening success, recent research has pivoted toward optimizing specific electrochemical properties via doping. Choy et al. applied six classical machine learning regression algorithms to predict the performance and optimize the composition of lithium nickel cobalt manganese (NCM) lithium-ion battery (LIB) cathode materials [71]. The study analyzed the correlation between the structure and elemental characteristics of 168 doped NCM systems, focusing on the initial discharge capacity and the discharge capacity after the 50th cycle (Fig. 7 (b)). Among the models, the Gradient Boosting (GB) model exhibited the best prediction performance for both IC and FC, with RMSE of 16.66 mAh g^{-1} and 18.59 mAh g^{-1}, respectively. Fig. 7 (c) illustrates the prediction performance of the GB model and the Random Forest (RF) model. It can be seen that the correlation coefficient scores from the GBM - IC ($R^2 = 0.76$) and GBM - EC ($R^2 = 0.64$) models both exceed the benchmark value of $R2 > 0.6$, which is used to consider a model as predictive. These high correlation values indicate that structural and elemental parameters such as lattice size and the radius of doped ions can accurately predict the discharge capacity of layered doped NCM cathodes. Variable importance analysis revealed that NCM materials with higher lithium doping levels, lower dopant contents, and lower dopant electronegativity demonstrated higher IC and FC values. This approach effectively reduced the experimental trial-and-error scope, accelerating the screening process for efficient cathode materials.

Furthermore, to tackle the complexity of multi-valent systems where data is even scarcer, Chen et al. addressed the scarcity of magnesium battery data by enhancing a crystal graph convolutional neural network (CGCNN) trained on multi-ion (Li, Mg, Na, etc.) datasets, achieving high-precision voltage prediction for Mg electrodes (MAE 0.29 V) [72]. As illustrated in Fig. 7 (d), the modified CGCNN framework represents crystals as multigraphs, with atoms embedded into nodes and interatomic connections encoded as edges. The model selectively pools features from active ions to predict electrode voltages. By screening 15,308 candidate materials from the Materials Project and GNoME databases, 160 high-voltage (>3.0 V) and high-volumetric-capacity (>800 mAh/cm^3) structures were identified. Subsequent machine learning molecular dynamics (MLMD) simulations using Neural Equivariant Interatomic Potentials (NequIP) assessed Mg-ion conductivity, narrowing the selection to 23 candidates with both high energy density and ionic conductivity (>100 mS/cm). Visualization analyses (e.g., t-SNE and PCA) confirmed the model's ability to interpret material chemical environments, establishing an efficient and scalable paradigm for accelerating the discovery of multivalent ion battery materials. This progression illustrates AI's growing capability to handle the subtle structure-property relationships in disordered inorganic systems.

4.2. Anode materials

In the field of battery materials research, anode materials play a critical role in determining the overall performance of batteries. The compositions of anode materials are diverse (Fig. 8 (e)), including graphite-silicon composites, metal alloys such as tin, antimony, and lead combined with inert matrices to enhance stability and conductivity, nanosilicon-aluminum-iron alloys with optional additional elements, lithium iron sulfate fluorides with high capacity and stability synthesized via solid-phase reactions, and active materials embedded in hard carbon (such as silicon and tin oxides synthesized via solvothermal

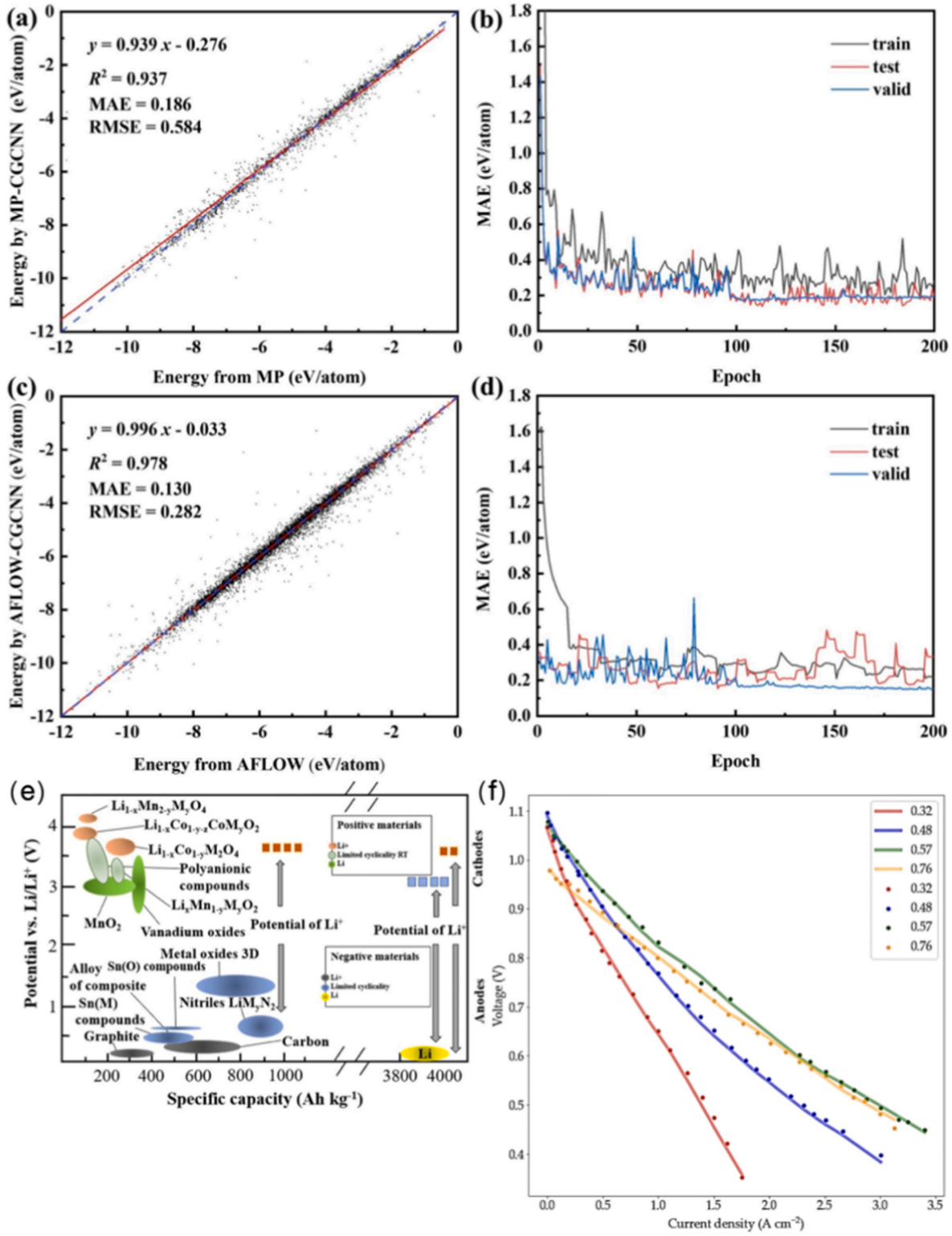


Fig. 8. (a–d) Performance of the CGCNN model in predicting the formation energy with data from two databases (MP and AFLOW) [81]. (a) Scatter plot of Gibbs free energy performance on CGCNN from MP database. (b) Plot of MAE with trained Epoch for training, test, and validation sets. (c) Scatter plot of Gibbs free energy performance on CGCNN from AFLOW database. (d) Plot of MAE with trained Epoch for training, test and validation sets. (e) Anode materials using in LIBs [73]. Copyright 2024, Cleaner Energy Systems(f)MLPs prediction and experimental data of cell current–voltage dependency at different anode Support Layer porosity (at 800 °C) [82]. Copyright 2022, MDPI.

methods) [73–75]. These diverse anode materials offer various advantages, such as high specific capacity, cost-effectiveness, environmental friendliness, and good stability, thus driving the advancement of lithium-ion battery technology [76,77].

Machine learning has been widely applied in the development of anode materials, bringing new insights into material development by enabling precise modeling and prediction of material performance [78]. Studies have shown that machine learning models, such as artificial

neural networks, Gaussian approximation potential energy, and spectral neighbor analysis potentials, can accurately predict the energy density and redox potential of anode materials [79]. These models facilitate the rapid screening of new materials and the discovery of candidates with high energy density and excellent electrochemical performance. By incorporating machine learning into the material design and evaluation process, researchers can efficiently explore the vast chemical and structural spaces of anode materials, thereby advancing the development of high-performance batteries [78,79]. Strategies for the development of inorganic anode materials include reducing particle size, controlling nanostructures, improving conductivity, and designing composites for materials such as FeO, Fe_2O_3, and Fe_3O_4 [80].

Shi et al. developed a machine learning approach based on the Crystal Graph Convolutional Neural Network (CGCNN) to efficiently screen binary alloy anode materials for lithium, sodium, potassium, zinc, magnesium, calcium, and aluminum batteries [81]. By integrating structural and energy data from tens of thousands of alloys in the Materials Project and AFLOW databases, the study identified candidates with low electrochemical potential (to suppress dendritic growth) and high specific capacity through predictions of formation energy, voltage, and capacity. Fig. 8 (a - d) illustrates the performance of the CGCNN model in predicting formation energies using data from both Materials Project and AFLOW. Parity plots revealed strong correlations between model-predicted and database-derived energy values, with AFLOW-trained models achieving a higher coefficient of determination ($R^2 = 0.978$) compared to MP-trained models ($R^2 = 0.937$), attributed to AFLOW's larger and more diverse dataset. Without experimental voltage inputs, the model validated over 30 experimentally recognized alloy systemsand predicted over 100 unexplored candidates with low potential (<0.3 V) and high capacity (e.g., Li alloys exceeding 600 mAh/g), such as Li-Mg and Na-Ca systems. The work also revealed design principles for alloy formation, such as alkali metals preferentially combining with Si or Sn. This framework accelerates the discovery of high-performance battery anodes, with future potential to expand material design by integrating generative AI tools like GNoME.

Golbabaei et al. employed machine learning approaches to predict the current-voltage relationship in anode-supported solid oxide fuel cells (SOFCs), utilizing a dataset comprising 403 experimental entries that encompassed operational parameters such as temperature, anode porosity, layer thicknesses, and current density [82]. The study evaluated the performance of multiple models, highlighting the superiority of a multilayer perceptron (MLP) neural network, which achieved a test R^2 of 0.998 and a mean squared error of 9.6×10^{-5}, outperforming conventional mathematical models and Gaussian process regression ($R^2 = 0.996$). The application of diverse machine learning algorithms demonstrated enhanced accuracy and reduced computational costs. Notably, the proposed MLP model predicted the influence of anode porosity on current-voltage dependency with percentage errors below 1 % (as illustrated in Fig. 8 (f)), while parameter variations exhibited reasonable consistency with prior literature. This work underscores the efficacy of machine learning in modeling complex energy systems, offering a data-driven strategy to minimize experimental efforts and optimize fuel cell design.

ThankGod et al. employed a high-throughput computational approach based on DFT combined with machine learning techniques to predict the theoretical voltage and capacity of $Ge_{12}C_{12}$ and $Si_{12}C_{12}$ nanocages [72]. By using algorithms such as linear regression and Lasso regression, they successfully predicted the voltage of anode materials from small sample data, achieving an R^2 value as high as 0.99, significantly improving simulation efficiency. Jiang et al. developed an effective machine learning classification model that integrates seven dopant characteristics and three electrochemical experimental features [61]. The model aims to predict the capacity of metal-ion doped TiO_2 anode materials in lithium-ion batteries (LIBs) using data science methods, ensuring the model results are intuitive and comprehensible. After comparative evaluation, the Gradient Boosting classification model showed excellent performance on the test set, with a true positive rate of 0.75, true negative rate of 0.92, and accuracy of 0.75, outperforming other models. Furthermore, the use of SHapley Additive Explanations (SHAP) method to explore feature importance revealed deep connections between dopant characteristics and capacity performance, identifying electronegativity and molecular weight as key features for capacity prediction.

4.3. Solid-state electrolytes

Solid-state electrolytes (SSEs) have garnered extensive attention due to their enhanced safety and exceptional energy density [83–88]. SSEs are generally classified into three categories: inorganic, polymer, and composite electrolytes. Fig. 9 (a) illustrates the distinct properties of these material classes. Inorganic electrolytes—typically comprising oxides, sulfides, and phosphates—exhibit high ionic conductivity and outstanding thermal stability, but suffer from pronounced brittleness and limited flexibility. Polymer electrolytes, based on a polymeric matrix, offer excellent mechanical compliance and versatile processability, yet their ionic conductivities are generally lower than those of inorganic counterparts. Composite electrolytes merge inorganic and polymer components to synergistically harness the advantages of both systems, aiming to balance high ionic conductivity with mechanical adaptability. Ionic conductivity is a critical parameter in SSEs, dictating the efficiency of ion transport and thereby directly influencing the performance of electrochemical devices such as batteries and fuel cells. The mechanisms of ion migration depend on the material class: in inorganic electrolytes, ions migrate via vacancy, interstitial, and interstitialcy pathways. Unlike polymer electrolytes driven by segmental motion, transport in inorganic systems is governed by crystal lattice defects. Consequently, applying AI to inorganic SSEs requires distinct strategies focused on predicting defect formation energies and ion migration barriers, as explicitly illustrated in Fig. 9 (b). Building on these mechanisms, future SSE development will concentrate on five key directions: (i) enhancing ionic conductivity to improve device performance; (ii) reinforcing high--temperature stability for reliable operation; (iii) engineering intelligent self-healing functionalities to extend cycle life; (iv) developing cost-effective fabrication processes to accelerate commercialization; and (v) expanding multifunctionality (e.g., dual ion–electron conduction) to broaden application scopes. Through the integration of machine-learning (ML) techniques with experimental and computational research, SSEs are poised to achieve breakthroughs in efficiency, stability, and sustainability, thus opening new avenues for electrochemical energy storage systems. Fig. 9 (c) presents the comprehensive ML workflow for predicting SSE properties, which—consistent with most ML applications—comprises four stages: data collection, feature engineering, model development, and model evaluation and deployment. Fig. 9 (d) highlights the principal targets of ML-assisted SSE design: precise tuning of interfacial stability to ensure electrode compatibility; prediction of mechanical strength to withstand applied stresses; optimization of thermal stability to resist temperature-induced degradation; and identification of optimal ion-transport pathways to enhance conductivity. By integrating these attributes, ML enables rapid screening and rational design of SSE materials, thereby accelerating the development of high-performance solid-state batteries and bridging the gap between theoretical prediction and practical application.

Cubuk et al. proposed a transfer learning-based method for the rapid screening of solid-state lithium-ion conductor materials using small datasets [90]. Traditional machine learning methods face difficulties in effectively applying to small datasets due to the lack of physical constraints. However, by incorporating descriptors guided by physical models, the predictive power of the model can be significantly improved. The first step in the transfer learning framework is to train a high-accuracy model based on physical descriptors, achieving a validation accuracy of 90 % on an experimental dataset (exp-dataset). Subsequently, a model based on general descriptors is trained using the

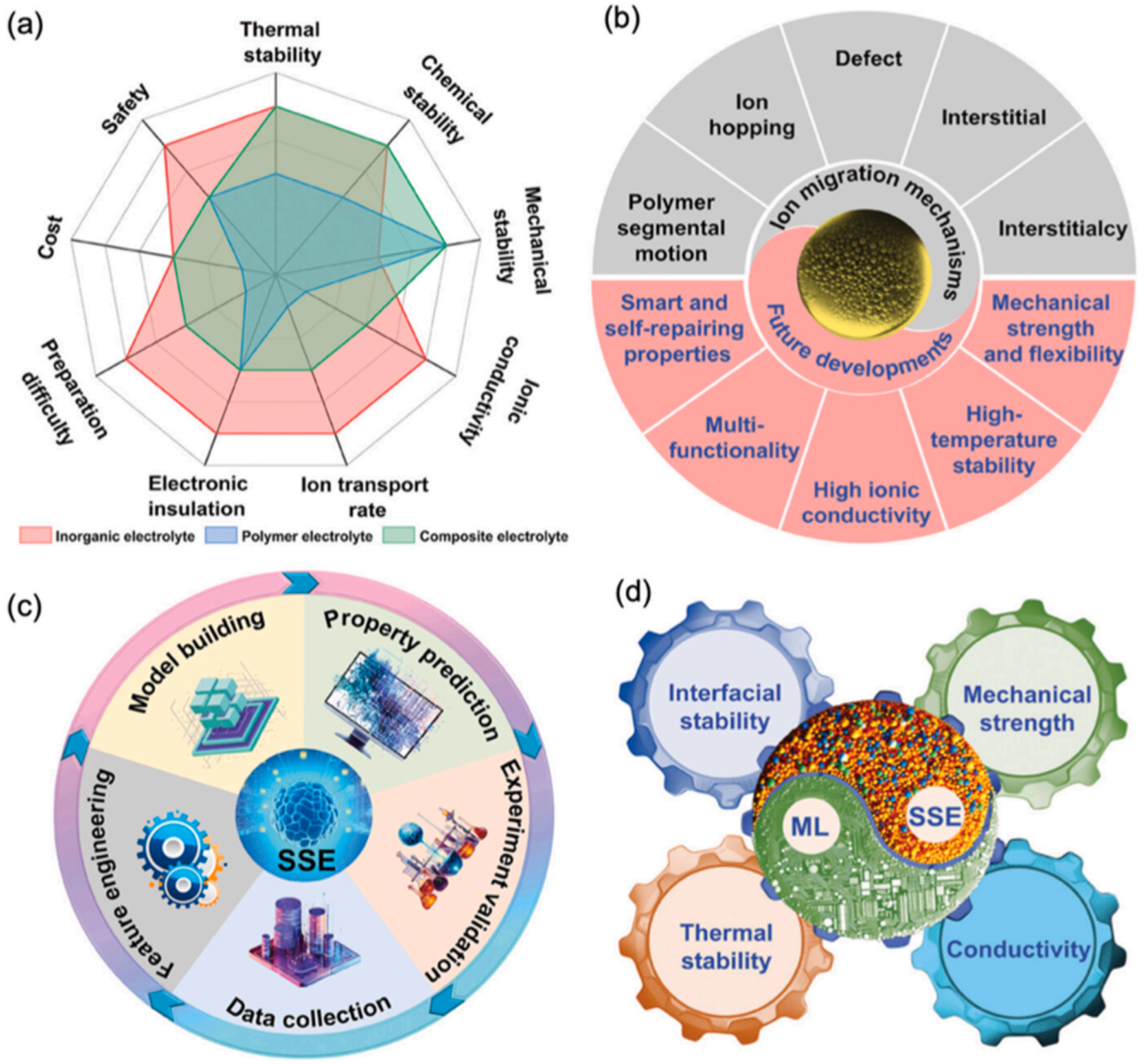


Fig. 9. SSEs Performance and ML prediction. (a) Radar plots for the properties of varying SSEs. (b) The ion migration mechanisms and future developments of SSEs. (c) The process of using ML to predict the properties of SSE. (d) ML-assisted prediction of SSE properties and design of SSE materials [89]. Copyright 2024, Advanced Energy Materials.

predictions generated from the physical model, enabling generalization to a larger-scale dataset. Ultimately, this transfer learning method screened 1392 potential high-conductivity candidate materials from a set of 12,716 materials and further filtered approximately 7 % of materials as high-performance lithium-ion conductors from a virtual combinatorial library of 2 billion compounds.

Research in electrolytes has traditionally focused on predicting ionic conductivity, but the frontier is shifting toward the complex challenge of interfacial stability and formulation design. In the domain of conductivity prediction, supervised learning has proven highly effective. Fujimura et al. combined experimental data with ab initio molecular dynamics (AIMD) calculations and used a SVR model to predict the ion conductivity of the $Li_{8-c}A_aB_bO_4$ ($A = Zn^{2+}$, Mg^{2+}, Al^{3+}, Ga^{3+}; $B = Si^{4+}$ or Ge^{4+}) system, significantly improving design efficiency [91]. Sendek et al. analyzed 39 experimental datasets using machine learning and calculated 20 descriptors based on atomic mass, electronegativity, and structural parameters. They successfully predicted the ion conductivity of various solid-state electrolytes, discovering materials such as $Li_5B_7S_{13}$, which exhibited an ion conductivity of up to 0.074 S/cm [92]. Xu et al. developed a logistic regression model to distinguish between high and low ion conductivity (greater than or less than 10^{-6} S cm^{-1}) based on 70 sets of LISICON/NASICON ion conductivity data [93]. This model used simple elemental characteristics as features and achieved an accuracy of 84 %, enabling the rapid screening of high ion conductivity LISICON/NASICON materials. However, predicting conductivity alone is insufficient if the material is unstable or the formulation is suboptimal.

To address the multidimensional challenge of stability and formulation, recent studies in 2024 and 2025 have adopted unsupervised learning and clustering strategies to map the chemical space. In a breakthrough study, Huang et al. utilized AI-driven K-means clustering to analyze molecular orbital energies and solvation energies. By grouping solvents and salts based on these descriptors, they identified a unique “delocalized” electrolyte design that breaks the trade-off between local stability and transport [94]. This AI-guided design enabled a lithium metal pouch cell to achieve a record energy density of 604.2 Wh kg^{-1}. Similarly, in aqueous systems where the electrolyte stability window is narrow, Li et al. combined DFT calculations with Machine Learning (SISSO algorithm) and AI clustering to screen electrolyte additives [95]. By using surface free energy as a stability descriptor, the model successfully identified 1,2,3-butanetriol as an optimal additive, which was experimentally verified to significantly enhance cycling stability. These latest works signify a paradigm shift: AI is moving

beyond predicting single scalar properties (like conductivity) to designing complex, multi-component chemical environments.

Similar to ion conductivity, the ion migration energy barrier is another critical ion transport property in solid-state electrolytes, reflecting the variation of ion conductivity with temperature. Zhao et al. proposed an machine learning method to predict the lithium ion migration energy barrier, achieving a RMSE of only 0.02 eV [96]. They focused on silver germanium sulfide and constructed 50 solid-state electrolyte structures, further calculating the lithium ion migration energy barrier using the bond-valence sum energy (BVSE) method and establishing an machine learning dataset. Additionally, a hierarchical decoding method based on crystal structures was employed to select features, including ionic radius, lithium content, unit cell volume, lithium-lithium atomic distance, bottleneck size of lithium diffusion path, configurational entropy, and ion occupancy. Using partial least squares regression, a quantitative relationship was constructed to predict the lithium ion migration energy barrier.

Honrao et al. developed an interpretable machine learning model capable of effectively predicting various physicochemical properties of materials (e.g., lithium ion migration barriers, redox potentials), thereby facilitating the high-throughput screening of solid-state electrolytes or anode coating materials [97]. In their work, the lithium ion migration barrier model included 22 structure-related features, such as lithium atom ratio, average number of neighboring lithium atoms, and average lithium atom distance. Moreover, Shapley values and permutation feature importance analysis were employed to assess the interpretability of the machine learning model, identifying key factors that influence the model's predictions. Through high-throughput screening of over 15,000 materials, potential solid-state electrolytes or anode coating materials, such as Li_9S_3N, $LiAlB_2O_5$, $LiYO_2$, $LiSbF_4$, and $Sr_4Li(BN_2)_3$, were identified. Key transport properties, such as electronic conductivity and ion conductivity, can serve as references for researchers, and machine learning methods can be used to screen large-scale materials through high-throughput calculations to identify potential solid-state electrolyte materials for lithium-ion batteries.

In summary, AI techniques have played a pivotal role in the research of inorganic battery materials, including cathodes, anodes, and solid-state electrolytes. These technologies provide powerful tools and novel approaches for performance optimization, discovery, and design of new materials, and they are expected to drive further breakthroughs in battery technology in terms of energy density, power density, cycle life, and safety.!

5. Conclusion and prospects

The rapid development of AI, particularly machine learning and deep learning technologies, has brought revolutionary changes to the design and performance prediction of energy storage materials. By efficiently mining experimental and computational data to establish structure–property relationships in complex systems, machine learning is increasingly becoming an essential tool for accelerating the development of lithium-ion battery materials. Compared with traditional experimental and computational methods, machine learning not only significantly reduces resource and time costs, but also provides material scientists with profound insights into the physicochemical properties, aiding in precise design and efficient screening.

In the research of cathodes, anodes, and electrolyte materials, machine learning models have been widely applied to predict material properties, optimize compositional ratios, and design new structures. For example, in the case of lithium nickel cobalt manganese oxide (NCM) cathode materials, gradient boosting models and SVR models have successfully predicted their initial discharge capacity and capacity retention after multiple cycles. These data-driven studies reveal intrinsic patterns in material performance by extracting key feature variables, offering a simple and effective prediction pathway for complex systems. Additionally, the combination of machine learning with DFT and molecular dynamics (MD) simulations has further enhanced our understanding of the intrinsic properties and dynamic behavior of materials. For instance, in the study of solid-state electrolytes and two-dimensional materials (e.g., MXenes), data-driven high-throughput screening and model optimization have significantly accelerated the discovery process of high-performance energy storage materials.

Despite the transformative potential of AI, its practical deployment in inorganic battery research is impeded by four fundamental challenges [96]: Data Sparsity and Survival Bias: Unlike the "Big Data" in computer vision, battery materials suffer from "Small Data." More critically, literature is dominated by positive results, creating a "survival bias" where models never learn from failed experiments. Furthermore, experimental data is highly sensitive to inconsistent metadata (e.g., trace water content, varying C-rates), leading to models that fail to generalize across different laboratories. The "Synthesis Gap" (Realizability): There is a persistent disconnect between thermodynamic stability (calculated by DFT) and kinetic synthesizability. Current generative models often "dream up" crystal structures that minimize energy but are impossible to synthesize due to complex phase competition or undefined reaction pathways. Predicting what to make is solved; predicting how to make it remains an open challenge. Interpretability vs. Accuracy Trade-off: Deep learning models, particularly Graph Neural Networks (GNNs), achieve high accuracy but often function as opaque "black boxes." In materials science, predicting performance is insufficient; researchers require causal interpretability—understanding which structural motifs drive ionic conductivity or stability—to derive universal design principles. The Multi-Scale Gap: Most AI models operate at the Ångström scale (atomic properties), yet battery failure occurs at the macroscopic scale (e.g., particle cracking, thermal runaway). Integrating information across these scales is difficult because key mesoscopic modalities (such as microstructure evolution and pore distribution) are rarely quantified in standard datasets, creating a "missing link" in performance prediction.

To overcome these challenges, a hybrid "data + mechanism" paradigm is emerging that marries statistical learning with domain physics. By embedding prior knowledge and physical constraints into ML architectures, researchers have built hybrid models that generalize better and yield more interpretable predictions [98]. For example, Physics-Informed Neural Networks (PINNs) incorporate electrochemical transport equations into the loss function; such models have accurately predicted battery state of charge and state of health even with limited training data. Likewise, multimodal fusion techniques explicitly combine disparate data streams – structural images, spectroscopic signals, and charge–discharge profiles – into unified models that capture complementary information [99]. At the same time, federated and privacy-preserving learning frameworks are being developed to break down institutional data silos: multiple laboratories can collaboratively train shared models on their local datasets without revealing sensitive raw data [100]. Together, these advances leverage physics and cross-institutional knowledge to produce AI models that are more robust, adaptive, and explainable.

Looking forward, the evolution of AI in inorganic battery materials is poised to expand beyond closed-loop optimization into broader, transformative directions. First, the emergence of "Foundation Models" or "Large Atomic Models" represents a paradigm shift from task-specific algorithms to universal potentials. Unlike traditional models trained on narrow datasets, these foundation models, pre-trained on vast chemical spaces (e.g., millions of DFT relaxations), can generalize across the periodic table. This capability will enable researchers to simulate complex interfaces, defects, and degradation mechanisms in solid-state electrolytes or cathode coatings with ab initio accuracy but at varying scales, significantly mitigating the data scarcity issue in specific battery systems.

Second, AI is transitioning the workflow from "high-throughput screening" to "generative inverse design." While current approaches predominantly filter existing databases, future generative models—such

as diffusion models and flow matching networks—will act as creative engines. By conditioning on desired performance metrics (e.g., specific ionic conductivity thresholds or voltage windows), these algorithms can explore the vast, uncharted chemical space to "dream up" novel metastable structures that do not exist in current repositories, effectively turning the material discovery process from a search problem into a generation problem.

Third, bridging the gap between theoretical prediction and experimental realization remains a critical frontier. Predicting a high-performance crystal structure is futile if it cannot be synthesized. Future AI research must integrate "synthesizability" as a core optimization parameter. This involves developing NLP-based agents capable of mining synthesis recipes from centuries of literature and using thermodynamic modeling to recommend precise precursors and sintering conditions. The integration of such synthesis-aware AI with robotic platforms will be key to translating digital predictions into physical battery prototypes.

Finally, a hybrid data–mechanism paradigm—synergizing foundation models, generative design, and autonomous synthesis—will herald a new "fourth paradigm" of materials science, enabling sustainable, high-performance battery technologies at unprecedented speed.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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